Defense Script EN

19 principal slides · 600 seconds · 14 appendices

1. Temporal-proxy classification with sEMG

Scheduled time: 20 s.

This study classifies an early-versus-late proxy created from each recording’s position in time. The ZIP does not contain independent measurements to claim true muscle-fatigue detection. I will describe the corpus, signal, pre-analysis, preprocessing, models, and the auxiliary experiment, separating completed results from the proposed design.

Source: Integrative biomechanics project.

2. Corpus and quantitative overview

Scheduled time: 30 s.

The main ZIP contains 140 CSV files and 4,256,257 samples. The 138 R1 to R5 recordings enter formal processing; two R0 files remain controls. Fifteen groups come from folder organization. Both legs and every series stay together during grouped evaluation. The RAR remains separate because participant identity and cross-archive links are not verified.

Source: inventory.csv and coverage.csv.

3. Recorded signal and time diagnostics

Scheduled time: 30 s.

The CSV files contain relative time, ADC counts and an exporter column computed as round(ADC_Value × 3300/4095, 2). That arithmetic conversion does not verify physical calibration, channel mode, offset or gain. The plotted trace uses one selected QC-valid run, while processing preserves every sample. We inspected clock gaps, jitter, constant segments and repeated extremes.

Source: Original CSV, private source audit and preprocessing.py.

4. Quantitative signal-quality pre-analysis

Scheduled time: 30 s.

The summary reports contiguous-run QC, uncalibrated ADC amplitude summaries and the number of recordings whose 50 or 60 Hz line prominence exceeds the 10 dB notch trigger. No synchronized repetition or movement-phase references are present, so those feature trajectories are unavailable.

Source: qc.csv, line_prominence.csv and preanalysis.py.

5. Signal preprocessing

Scheduled time: 40 s.

The historical pipeline separates signal at acquisition gaps and rejects excessive clock jitter and constant segments. It uses a fourth-order Butterworth low-pass prototype transformed into a digital 20 to 400 hertz band-pass, applied bidirectionally in second-order sections. One hundred milliseconds at each edge are masked. The optional notch is not in the primary pipeline. Processing is offline.

Source: configs/study.yaml, qc.py and preprocessing.py.

6. Windows and temporal proxy labels

Scheduled time: 40 s.

Complete windows in the first elapsed-time third receive proxy class zero; windows in the final third receive class one. The central third and boundary-crossing windows are excluded. A candidate-activity criterion selects usable windows. The observation is a window; the learned target is temporal position, not measured muscle fatigue.

Source: label_audit.json and coverage.csv.

7. Ten features per window

Scheduled time: 30 s.

The input contains three amplitude features, three spectral features and four relative db4 wavelet energies. The two-stage proxy experiment and direct comparator use the same ten original features. Time, file identity, side and series are excluded. The scaler is learned only from training folds, and no MVC normalization is claimed.

Source: features.py and feature_schema.json.

8. Grouped validation method

Scheduled time: 40 s.

Each outer round holds out one entire recording-derived group. The other groups are used for grouped inner model selection. Scaling and fitting use training data only. Each eligible window receives an out-of-fold prediction. Uncertainty is summarized across groups rather than treating overlapping windows as independent subjects.

Source: evaluation.py, splits.json and evaluation_receipt.json.

9. Models and parameter selection

Scheduled time: 30 s.

The majority model is a trivial reference. Logistic regression on RMS and MDF is the simple baseline. Full logistic regression, an RBF support-vector machine and a random forest test more flexible mappings. Nested selection chooses a family inside grouped validation; it does not average model votes.

Source: evaluation.py and inner_search.csv.

10. Performance on the temporal proxy

Scheduled time: 40 s.

All procedures use the same 5,509 early/late windows and fifteen held-out groups. The nested selector has mean Accuracy 0.6628 and macro-F1 0.6240; logistic RMS plus MDF has 0.6535 and 0.6272. The paired Accuracy difference is +0.0093, 95 percent group-bootstrap interval [-0.0373, +0.0546]; the macro-F1 difference is -0.0032, interval [-0.0591, +0.0509]. Both intervals include zero. These are metrics of recording position, not physiological fatigue.

Source: evaluation/metrics_summary.csv and paired_differences.csv; accuracy_paired_differences.csv.

11. Main proxy errors

Scheduled time: 30 s.

The pooled confusion matrix shows errors for the early and late temporal classes. Group-level macro-F1 varies substantially, so pooled counts do not replace across-group uncertainty. Without independent references, these errors cannot be assigned a physiological cause or interpreted as fatigue mistakes.

Source: evaluation/oof_predictions.parquet and evaluation/results.csv.

12. Auxiliary EMG cascade

Scheduled time: 40 s.

The first stage is a random forest that classifies first versus second halves of candidate EMG bursts. Its score enters a logistic second stage together with the original ten features and selected interactions. The auxiliary label is an EMG burst half, not observed knee extension or flexion.

Source: phase_spike.py and the frozen exploratory protocol.

13. Auxiliary experiment design

Scheduled time: 30 s.

The variants use identical windows, temporal labels, scaling and held-out groups. We compare the direct ten-feature logistic model, EMG plus score, conditioned interactions, score only and a random-label control. The auxiliary head was fit on 3,337 windows from 996 candidate bursts.

Source: model_registry.csv and phase protocol.

14. Auxiliary performance on the same proxy

Scheduled time: 35 s.

For the temporal proxy, direct logistic reaches Accuracy 0.6752 and macro-F1 0.6355; the conditioned variant reaches 0.6701 and 0.6293. The paired Accuracy difference is -0.0051, 95 percent interval [-0.0222, +0.0095]; the macro-F1 difference is -0.0062, interval [-0.0232, +0.0082]. Both include zero. The result does not show an improvement on the temporal proxy and says nothing about true fatigue detection.

Source: phase_experiment/metrics_summary.csv, paired_differences.csv and accuracy_paired_differences.csv.

15. Targets for each cascade stage

Scheduled time: 30 s.

The auxiliary confusion matrix evaluates burst-half labels, while the second-stage matrix evaluates early-versus-late recording position. The auxiliary head has mean group Accuracy 0.5775 ± 0.0783 and macro-F1 0.5420 ± 0.1028. The conditioned second stage has Accuracy 0.6701 ± 0.1022 and macro-F1 0.6293 ± 0.1330 on its own support. The first matrix is not extension versus flexion, and neither is fatigue versus no fatigue.

Source: confusion_matrices.csv and auxiliary_phase_fold_metrics.csv.

16. Perceived-fatigue classification by aligned window

Scheduled time: 25 s.

This section addresses the corrected target. Version 2 of the external dataset has 13 participants, 156 trials and four sEMG channels. Every 500 millisecond window must lie wholly inside one stable self-report state. Stable 0 means no perceived fatigue; stable 2 means perceived fatigue; transition state 1 is excluded. Evaluation holds out each participant. The reference is self-report rather than force loss.

Source: Zenodo v2, DOI 10.5281/zenodo.14182446; fatigue-window architecture.

17. Per-window model results and confusion matrices

Scheduled time: 25 s.

The table reports participant-macro Accuracy and macro-F1 with participant-bootstrap 95 percent intervals for global HGB, nested-tuned HGB, prime-mover specialists and the time-only control. Confusion matrices show OOF counts and row-normalized rates. The nested procedure selects parameters using training participants only. The specialists route by known prime mover, not by knee extension or flexion. Time strongly outperforms EMG, showing temporal confounding and weak generalization.

Source: Benchmark metrics, nested HGB results, specialist comparison and confusion matrices.

18. Separate trial-level benchmark

Scheduled time: 25 s.

This dataset has a different target: one outcome label per complete trial. It compares a reference contraction with TLim sustained to task failure. Accuracy from a perfect time/bout control is a protocol shortcut because the classes also differ in active-bout structure. This result is not a per-window fatigue classifier and contains no extension/flexion phase label.

Source: Figshare task-level benchmark, DOI 10.6084/m9.figshare.15273906.v1.

19. Conclusions

Scheduled time: 30 s.

The original course ZIP model predicts an early/late proxy. A separate external experiment does train on aligned self-reported perceived-fatigue labels per window, but EMG performance is weak against the time-only control. The phase cascade and extension/flexion experts cannot be trained with the available phase labels, and neither external dataset supplies objective force-loss labels for this course cohort. The trial-level benchmark remains separate and exposes a protocol shortcut.

Source: Course proxy, perceived-fatigue benchmark and task-level benchmark.

ANEXOS / APPENDICES

20. Appendix: relationships among the models

Appendix for questions.

The project includes six principal procedures, five supplemental variants, an auxiliary head and a separate full-cohort fitted artifact. The matched ten-feature logistic baseline is distinct from the nested selection procedure. These roles should not be counted as independent evidence or independent cohorts.

Source: model_registry.csv.

21. Appendix: sensitivities and ablations

Appendix for questions.

Prespecified sensitivities and feature-family ablations test how proxy results change under alternate analysis choices. Some sensitivities use different retained windows. We report these scenarios as sensitivity checks and do not select the most favorable one as the headline result.

Source: sensitivity_summary.csv and ablation_summary.csv.

22. Appendix: principal confusion counts

Appendix for questions.

These are pooled out-of-fold confusion counts for each principal procedure on the same 5,509 windows. Class zero means the first recording third; class one means the last third.

Source: Primary confusion matrices.

23. Appendix: normalized principal matrices

Appendix for questions.

Each row is normalized within its observed temporal-proxy class. The matrices show class-specific error proportions, not validated physiological sensitivity or specificity.

Source: Normalized primary confusion matrices.

24. Appendix: auxiliary confusion counts

Appendix for questions.

These matrices compare the supplemental stage-two variants on the same early/late proxy support. The auxiliary burst-half head is shown separately because it has another target.

Source: Supplemental confusion matrices.

25. Appendix: normalized auxiliary matrices

Appendix for questions.

These row-normalized matrices show error proportions for each supplemental procedure. The first-stage head predicts EMG burst halves, not knee movement phase.

Source: Normalized supplemental confusion matrices.

26. Appendix: references for movement phase

Appendix for questions.

The cited literature motivates measuring movement phase and controlling kinematics when interpreting EMG. Those studies do not validate our burst-half labels or demonstrate improved fatigue prediction from this architecture.

Source: docs/PHASE_REFERENCES.md.

27. Appendix: window lengths and overlap

Appendix for questions.

The historical analysis uses 250 millisecond windows with 50 percent overlap. A separate study with independent references would use 500 milliseconds as its primary window and 250 milliseconds as a sensitivity analysis. At one kilohertz these yield nominal bin spacings of two and four hertz. Longer windows improve nominal frequency spacing; overlap increases temporal sampling but creates dependent observations.

Source: Frozen historical configuration and configs/reference_study.yaml.

28. Appendix: candidate feature formulas and audit

Appendix for questions.

Five candidate features were computed on 9,289 QC-valid active windows without reading labels or fitting a model. Integrated EMG and variance were perfectly rank-correlated with MAV and RMS under fixed duration. This is a descriptive redundancy result, not evidence of predictive value. The candidates were not added to the classifier; independent reference labels are required before testing Accuracy or macro-F1.

Source: docs/CANDIDATE_FEATURE_AUDIT_2026-11-09.md; granular outputs remain private.

29. Appendix: filter and window equations

Appendix for questions.

The fourth-order Butterworth analog prototype is transformed to a digital band-pass of order eight and implemented as second-order sections. Coefficients depend on verified sampling rate.
Forward-backward filtering is offline. The optional notch is used only if line-noise diagnostics justify it.
The historical spectrum uses a one-sided Hann periodogram, not Welch averaging.

Source: SciPy signal documentation: https://docs.scipy.org/doc/scipy/reference/signal.html.

30. Appendix: feature equations

Appendix for questions.

The ten input features summarize amplitude, spectrum and relative wavelet energy. The corrected waveform-length feature uses the interval between first and last samples and has a separate schema. It does not replace the frozen historical version or alter its results.

Source: features.py, feature_schema.json and docs/reference_fatigue_protocol.md.

31. Appendix: independent-reference architecture

Appendix for questions.

Synchronized angle or reviewed video provides extension and flexion supervision. Calibrated synchronized force or torque provides the approved fatigue label. The same feature matrix feeds the direct model, phase cascade and soft-gated experts. OOF phase probabilities are used in cascade training, and held-out participants are evaluated on matched support. This architecture has not been trained on the supplied ZIP.

Source: docs/reference_fatigue_protocol.md.

32. Appendix: label and model equations

Appendix for questions.

Phase labels are derived from angular velocity and an approved deadband. The candidate torque rule requires calibration, repeatability and approval. Direct, cascade and expert equations define three prospective models. The executed auxiliary formula still predicts EMG burst halves and a temporal proxy, not physiological fatigue. Fine-tuning would require compatible external labeled data; reinforcement learning is not suited to static window labels without sequential actions and rewards.

Source: docs/reference_fatigue_protocol.md and historical model registry.

33. Appendix: real signal and filter response

Appendix for questions.

The figure shows a one-second interval selected by the first continuous QC-valid window, without checking labels or model results. It includes centered stored ADC counts, SOS output, Hann periodograms before and after filtering, and the digital filter response at the observed rate. This is preprocessing evidence, not physiological performance or a fatigue class.

Source: Private signal_preprocessing_evidence receipt; anonymized figure.